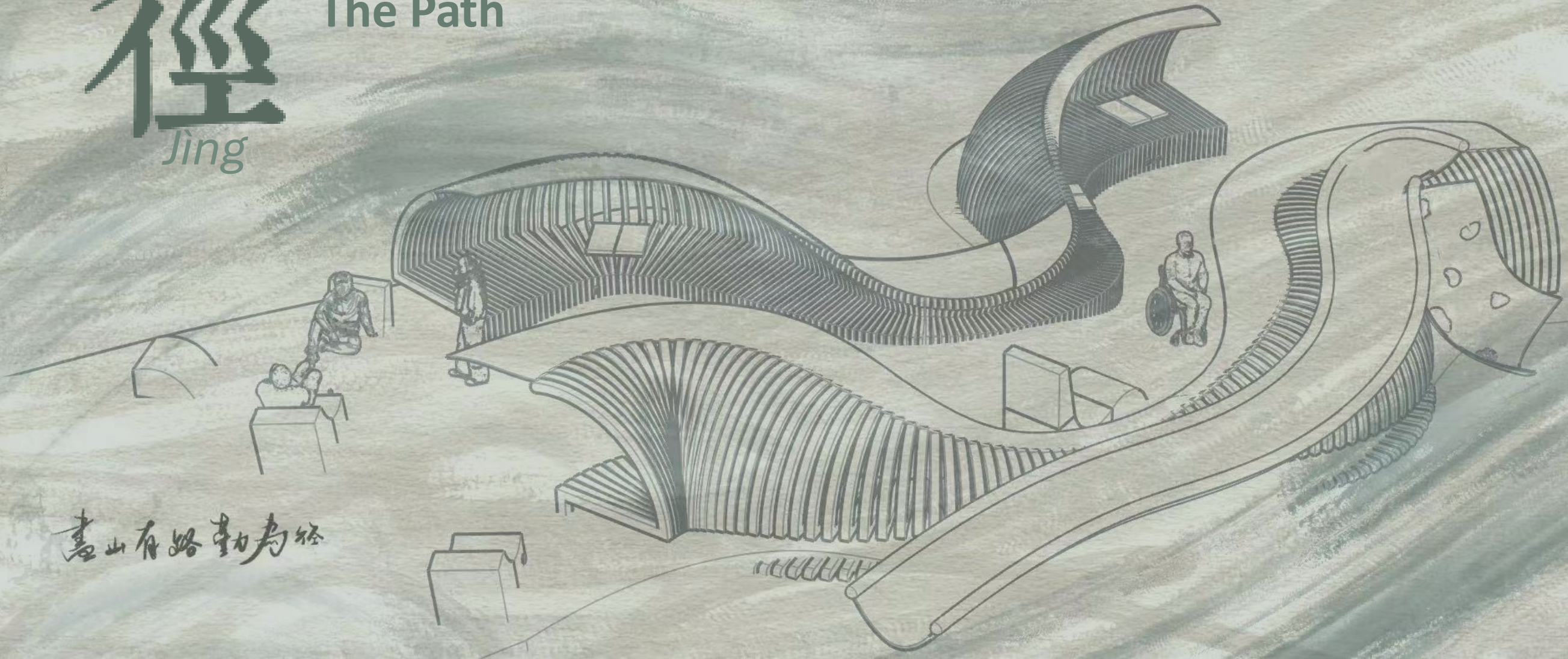


徑
Jìng

The Path



喜山有路勤為徑

"There is a path through the mountain of knowledge – and it is paved with diligence."

Bdobe Outdesign

Simon Xia, Yuxuan Zhou, Cici Song, Ziqi Zhou, Simon Wang

1. Problems & Context

Design Brief

Queen’s Lawn, the primary green space on campus, faces limited year-round usability due to **seasonal weather conditions**. The space **lacks sufficient seating, shelter, and social infrastructure**, reducing its adaptability and comfort throughout the year. Our project focuses on **creating a comfortable, attractive, playful, and multifunctional facility** for users to study, play, and relax in Queen’s Lawn and further increase **social interactions**.

Problem Area

Queen’s Lawn is the only central green space on campus, with strong potential to support student life. However, its usability is highly **dependent on seasonal weather**. In winter, the space becomes muddy, lacks sunlight, and is **rarely used** by students. In contrast, it is much more active in the summer, when conditions are more inviting. This seasonal disparity highlights key issues: a **lack of seating, shelter, and social infrastructure**, along with **poor adaptability to weather**. These limitations reduce its year-round appeal and suggest a need for simple, flexible interventions to improve comfort and functionality.

Map



Solutions?



Key insight

Consideration?

Given the space’s strong dependency on weather, the design should be **modular** and **adaptable** to seasonal changes. It’s important to consider elements like **sunlight** and **shading**; it should provide **shelter** when raining, while still allowing **natural light** on sunnier days. A flexible, short-term solution to improve comfort and usability throughout the year.

- A lack of seating
- Lack of school culture
- Aesthetics of the space
- Low Adaptability to Weather Conditions

Statistics

3,1000



Population

Area
1700 m^2

Number of
benches



20-50



Key Users



Problems

Primary User – Students

- Students who walk through the grassland account for the largest portion of the total population in Imperial.
- It is important to provide enough space and facilities for students to study and relax.



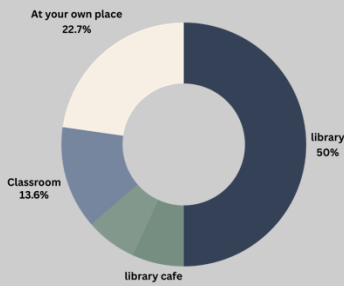
Secondary User – Professor & staff

- Although many Professors and staff are on campus, they use the space less frequently.
- From user research, many professors prefer to stay in their offices

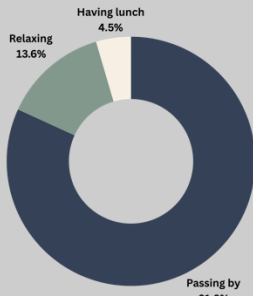


Survey

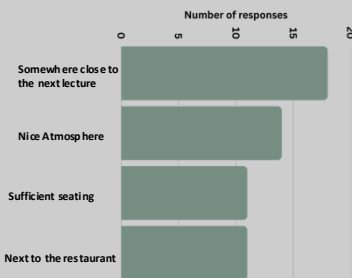
Where do you usually have lunch?



Why do you visit the Queen’s Lawn?



How do you decide where to eat/relax/study?



5Ws & 1H

Where
Queens’ Lawn, Imperial
College London



How
designing a **flexible**
structure that adapts
to **weather and**
student needs.

Who
Primary User:
Students
Secondary User:
Professors & Staff

When
All year round

Why
Motivate interactions
between people by
improving **space**
management and
providing more facilities

What
A loop combining seating
and shelter to create a
functional and **visually**
engaging space for rest,
study, and social
interaction on campus

2. User Research & Idea Development

Primary Research

Observation

Winter

- **Limited Seating:** Few places to sit, making eating uncomfortable.
- **Poor Grass Condition:** Grass is worn out in winter.
- **Crowded on Sunny Days:** Area gets busy when the weather is nice.



Summer

- **Sitting on Grass:** Many choose to sit directly on the lawn.
- **Poor Maintenance:** Untidy grass and flowers affect the atmosphere.

Interviews

Student

- **Convenient Spaces:** Students need more comfortable seating for daily use and comfort
- **Maintenance Issues:** Poorly kept, uninviting for students to use (Mainly winter)
- **Lack of Facilities:** More facilities such as seats and lights, to improve comfort

Professor

- **Underused Space:** The area is not fully utilised, limiting its value to student life.
- **Need for Cultural Link:** The redesign should reflect the university's values to strengthen identity and community.
- **Design Constraints:** Physical or institutional limits may affect redesign plans and should be addressed early.

Insights

Insufficient Seatings

Limited seating and **poor bench placement** reduce the comfort and usability for students.

Weather Condition

Lack of **waterproofing** and **limited indoor seating** leave students without usable space in rain or cold weather, reducing year-round comfort and usability.

Space Management

Good weather leads students to sit on the ground due to **limited outdoor options**, but **formal seating** is still needed to support **study and relaxation**, to support varied student needs.

Aesthetic

Users seek a more **pleasant** environment, but poor grass, lack of flowers, and **seasonal aesthetics** make the space feel dull. Rain further limits use. Adding **decorations** and **weather-resistant features** could enhance year-round appeal.

School Culture

Staff observed the space is **underused** and **disconnected** from the school's **hardworking culture**. Redesigning it to reflect this spirit can help make the space more meaningful for both students and visitors

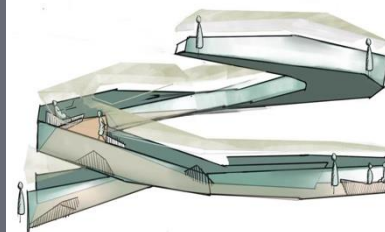
3 Concepts

InnoRest



A 360-degree **rotatable** bench designed to enhance **comfort, usability, and sustainability** in outdoor spaces. It features **solar-powered wireless charging panels** to support electronic devices, making it ideal for students and staff to rest and work. The backrest is **adjustable** for **ergonomic** support, and a **foldable table** is included for studying, dining, or casual work. Additionally, a **built-in screen** can display educational content, campus announcements, or event information, supporting both learning and communication needs in shared spaces.

The Path



This two-level **spiral** staircase offers elevated views and ample seating, enhancing both **functionality and user experience**. The structure is **partially covered**, allowing it to be used comfortably in various weather conditions. Seating is thoughtfully **integrated** along the staircase, utilizing the height differences to create unique resting and observation points. The design not only encourages movement but also maximizes space efficiency in outdoor environments.

HoloHub



The **glass** house is equipped with various facilities to provide seating and enhance user convenience. All lifts have been replaced with stairs to promote **physical activity** and create a more **open, accessible** interior layout. At the center, a holographic projection of the letter "**I**" serves as a symbolic focal point. The interior is further enriched with **planted trees and greenery** on the ground level, offering a **calm and natural** atmosphere for relaxation, study, and informal gatherings.

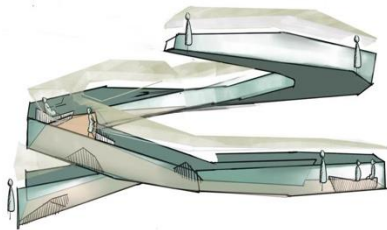
3. Concept Selection and Improvement

Key design consideration

- 1. Adaptability to different circumstances
- 2. Volume
- 3. Multifunctionality
- 4. Feasibility



InnoRest-Multifunctional Bench



The Path-Spiral Staircase

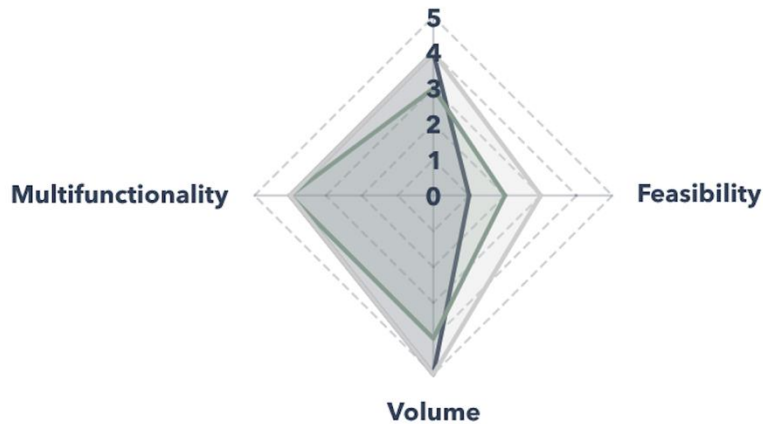


HoloHub: Glass House

a. Quantitative Analysis: Rating form1-5

Concept	Adaptability to different Circumstances	Volume	Multifunctionality	Feasibility
InnoRest	3	4	4	2
HoloHub	4	5	4	1
The Path	4	5	4	3

Adaptability to different circumstances



b. Student Feedback

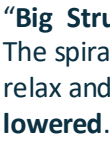
Students form18-20 are invited from imperial are invited to see our initial sketches and give feedbacks.



“The **staircase** design is brilliant. It doesn’t take up much ground space and still adds a lot of seating. Very **efficient**.”— Joe DE1



“But if it’s just **benches**, it might not be enough for **group activities**. It feels more suitable for short breaks.” --Cici Chem1



“**Big Structures** might be less feasible. The spiral staircase is a good design for relax and study. It would be better if it is **lowered**.” —Tony DE1



“I think a structural design might be more **interesting** for different uses.” —April ChemEng1



“Wouldn’t the **construction cost** of the **glass house** be quite high? The full-glass structure and projection system seem a bit complex.” — Sarah Earth Science 1



“The idea of walking or sitting around a **spiral staircase** gives off such a relaxing vibe. I’d rather read here than in the library.”— Edward BME

c. Validation of the reason of concepts

The selection of the Spiral Staircase is supported by several key advantages that align with both user needs and design goals. These strengths highlight its potential to enhance **functionality, comfort, aesthetics, and cultural relevance** within the space.

Advantages of The Path

- **Space-Efficient Seating:** Utilizes **vertical** space to address seating shortages, with integrated seating along the steps.
- **Weather-Resilient:** Partial covering improves **comfort** in varied weather, supporting year-round use.
- **Improved Aesthetics:** Adds **visual interest** and can be enhanced with seasonal decorations or greenery to uplift the currently dull environment.
- **Supports Study and Relaxation:** Offers both quiet spots for studying and open areas for group use, with a **relaxing, inviting** atmosphere.
- **Cultural Symbolism:** Reflects **ambition** and **movement**, reinforcing the school’s hardworking identity and public image.

As a result, the (Spiral Staircase) has been selected as the final concept due to its well-organized seating layout, weather-resistant design, and strong aesthetic appeal.

4. Prototyping Feature Development

The improved design is refined through **detailed sketches** and **CAD modelling**, clearly outlining its **functions** and **spatial dimensions**. Following this, a thorough **material investigation** was conducted to identify suitable options for the **modular seating components**. The final material selection will be informed by user surveys to ensure the choices align with students’ preferences for comfort, aesthetics, and usability in outdoor settings.

Improvements of Concept – The Path

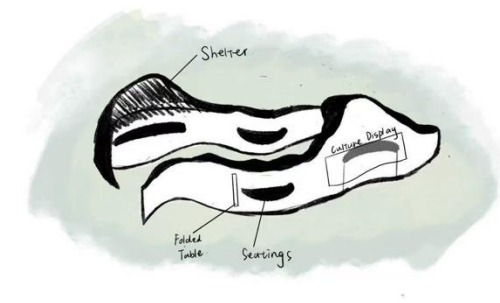
Problems

- **Accessibility and practicality concerns:** The original spiral staircase, though **visually striking**, was less **feasible** for construction and daily use. Reshaping it into a lowered infinity loop makes the design more **stable, accessible, and user-friendly**.
- **Seasonal Change:** The structure responds to **seasonal variations** in student behaviour. During hot and sunny days, students tend to prefer sitting under a shaded area. In contrast, during colder months with sunny weather, they are more inclined to seek out sunlit spots.

Improvements

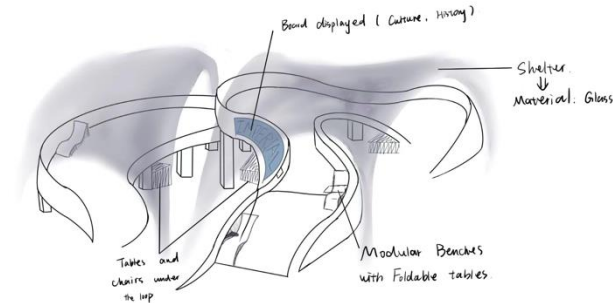
- ✓ **Modular seatings allow flexible transformation**, enabling the design to adapt to various arrangements depending on students’ preferences for **different purposes**. This **versatility** ensures the space remains **functional and welcoming** throughout the year, whether for **studying, resting**, or engaging in group activities.
- ✓ **The spiral staircase concept has been lowered and refined** into a structure resembling an infinity symbol. This new form improves **feasibility, accessibility, and structural stability** while retaining the aesthetic value and symbolic meaning of continuity and connection.
- ✓ **Sheltered areas are thoughtfully integrated** into the structure, protecting from both sunlight and rain. During hot weather, users can stay comfortably shaded beneath the pillars, while in colder conditions, they can enjoy the warmth of the sun on the platforms beneath the glass-covered roof.

Path



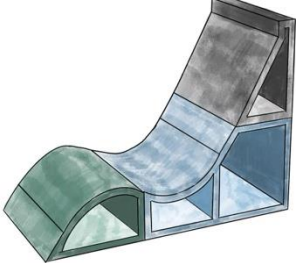
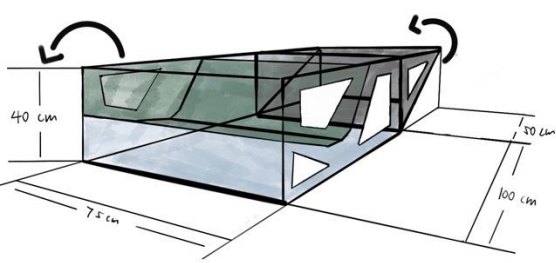
A 2-metre-high ground-level pathway features **modular seating** and **foldable tables** on both sides, allowing students to walk through and rest. **Imperial’s culture** is showcased along the side wall.

Loop



A loop structure elevated approximately 3 meters above the ground provides shelter, with **modular seating and foldable tables** strategically positioned along and beneath the loop. Moreover, something related to Imperial culture will be displayed on the side wall.

Key Features



Teak wood



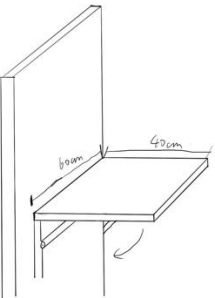
HDPE (high-density polyethylene)

Folded: Seats for 2-3

Expanded: Comfort for single

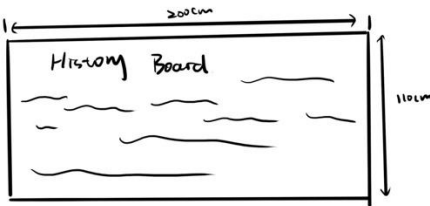
The **Modular Seating** is designed for different seating arrangements according to users’ preferences:

- **Movable & transformable:** movable and transformable chairs allows users to change the direction
- When **folded**, the structure accommodates the needs of multiple users simultaneously (2–3 users).
- When **expanded**, it transforms into a comfortable, ergonomically designed seating arrangement suitable for individual use only.
- **Material:**
 - Teak Wood: **Naturally weather-resistant, resists rot and insects, ages beautifully** (turns silvery gray)
 - HDPE (high-density polyethylene): **Lightweight, affordable, weather-resistant, and comes in various colours.**



Foldable tables are designed for:

- Designed for **versatility**, suitable for both studying and dining.
- **Easily foldable**, allowing for convenient storage and flexible use of space.
- Maximizes **space efficiency** while maintaining functionality and comfort.



History boards are designed for:

- Display the History of the school
- Link to **school culture**
- Improve **aesthetics**

Key Prototyping Questions

- What is the **shape** of the chair that makes it not only ergonomic but also satisfies the user's needs?
- What is the shape of the building to make it **weather-independent** and available for sufficient seating?
- How to **arrange** different components in the Structural Loop?

5. Prototyping Journey of Modular Seatings I

Prototyping Considerations (Based on Insights)

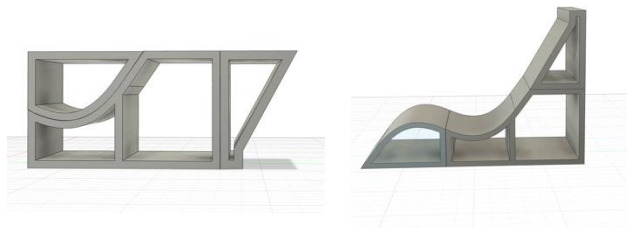
- Whether to add any **components** to satisfy more needs. Eg. Component to offer convenience in study
- **Shape**: Ensure user comfort through ergonomic design.
- Increase **seating capacity** to accommodate more users.
- How to make the chair **transformable**?

a. Prototype 1: Simple 3D printed Model

The modular seating was developed based on the structure presented in the previous slide. Clear **CAD models** (in Fusion 360) were created to illustrate the design, accompanied by photographs **showing users interacting** with the prototype, providing valuable insight into its functionality and user experience.

Folded: Seats for 2-3

Expanded: Comfort for single



This modular chair addresses the need for both **individual comfort** and **adequate seating** for multiple users (2–3 users) through its transformable design. It consists of three identical modules, with each module designed to have the same width to ensure smooth transformation between configurations.

Lying on the Chair

- ✓ **Posture**: Reclined or lying down.
- ✓ **Design Focus**:

Ergonomic: Supports natural body alignment while resting.

Individual Comfort: Meant for personal relaxation, reading, or resting.



Sitting on the Chair

- ✓ **Posture**: Upright or casually seated.
- ✓ **Design Focus**:

2–3 People: Accommodates small groups.

Semi-Formal Chatting: Encourages social interaction, informal meetings, or casual conversations.

Modular Support: Likely designed to adapt to multiple users.



pictures generate by ChatGPT 4o



The folding function in this prototype is achieved through the use of **adhesive tape**, which connects the corresponding edges of each module to allow for flexible transformation.

b. Prototype 2: Cardboard Model

After the shape is determined, a larger model that could be actually used in reality is made.

Modelling Technique

This is a to-scale model intended for real-world application. **Cardboard** was used to construct the full-sized prototype. To ensure the structure is stable enough to support a person weighing over **85 kg** while expanded and **180kg** when folded, **traditional Chinese mortise and tenon joints** were incorporated to reinforce the chair's form and structural integrity.



Cardboard Model

The Folding function was not incorporated into the current prototype of the chair. This decision allows for **flexibility** in future development and can be guided by user feedback, specifically whether users prefer the addition of a **hinge** for structured movement or find the unrestricted, hinge-free design more engaging and playful.



How to use it



The intended form has been successfully realized, and the prototype effectively accommodates both **seated use** for two individuals and a **lying posture** for a single user, demonstrating its multifunctional versatility.

Other ways of using

Maximum allow 3 people to sit



5. Prototyping Journey of Modular Seatings II

User feedback & Co-design activity

A co-design activity was conducted with **12 participants**, including **11 students and 1 professor**, to gather user feedback. Participants interacted with the 3D-printed model and tested the full-scale cardboard prototype by sitting on it. They were also asked to share their opinions on the chair's shape and comfort. In addition, the **necessity and practicality of foldability** were explored based on user responses.



Prototyping Questions

1. Is the shape **ergonomically designed** to provide a comfortable sitting experience when expended?

Users individually tested the seating and provided feedback regarding its **comfort and ergonomic design**. The adjustable front section allows individuals of varying heights to sit comfortably. **All 12 users** said is comfortable to sit.

2. Is **folding** really needed? Or flexible modular seatings are preferred.

Users were asked whether they preferred the folding feature or found it more **flexible** and engaging to use modular seats that can be separated for **different functional purposes**. 9 users said it is not necessary.

3. Does the design of seat Increase **seating capacity** to accommodate more users?

4. Should **additional components** be incorporated to better meet user needs—for example, features that support studying or enhance convenience?

Users' Comments

"The **top components** for lying should be connected to the body part so it's stable." —April, Chemical engineering

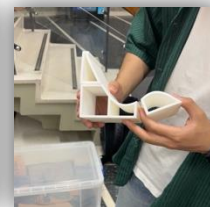
"The chair is **ergonomic and comfort** to me and I would like to use it if it is placed outside." —Joe, Design engineering

"It might be more **playful** to have the seating without the folding feature." —Josh, Design engineering

"Separated parts could be useful when I sit with my friends. This also increase the capacity." —Tony, Chemical engineering

"It's smart how the form uses space **efficiently**—more of us can sit together comfortably. The only thing is to add more seats like this." —Daisy, MBE

"Maybe you could add a **charging socket**—it would make things a lot more convenient." —Curtis, Design Engineering



For two



For three

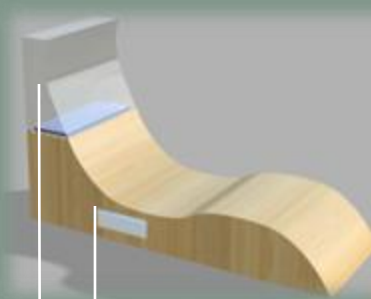


A charging socket is added on the right side of the seat for user convenience.

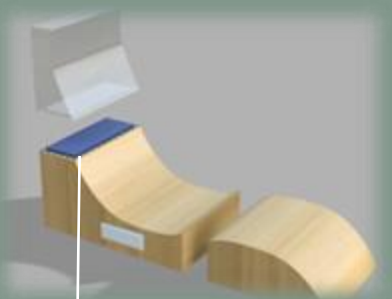
Things to improve

1. **Shape:** Good **ergonomic** modular seating, but make sure there's a connection between the **top component** for support of the head to be connected to the body.
2. **Folding:** **Folding functionality** is not considered essential, while separable components offer greater **versatility** in usage. Additionally, the connection between the top and the main structure should be **non-permanent and easily detachable**.
3. **Seating Capacity:** The modular structure already enhances **seating capacity**. It is recommended that the number of seats be increased to accommodate up to 35 users.
4. **Charging socket:** A **charging socket** will be added to charge electric devices, to provide a better environment for study. In addition, a battery source should be added.

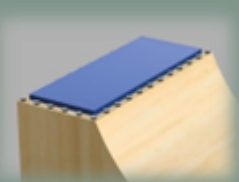
Prototype Iteration



Charging Socket



Solar Panel



An interlocking joint is used to connect the two part with help of magnets

The top component is made to be transparent to allow sunlight to get through to reach the solar panel to provide electricity for charging.

6. Prototyping Journey of the Large Structure

Making Prototypes



Prototypes of both the loop and path structures were constructed using foam and cardboard to visualize their spatial arrangements. The path model was built at a scale of **1 meters to 4.7 centimeters**, while the loop model followed a scale of **1 meters to 2.5 centimeters**. These scaled prototypes assist in exploring proportion, layout, and potential user interaction with the designs.

Loop



A large permanent structure of two floors

Path



A lower structure separating groups by its shape

User Feedback & Co-design Activity

The model was built on a scaled base representing Queen’s Lawn. Yellow tape was used to represent the central pathway. Additionally, **3D-printed scale figures and chairs** were placed on the model, allowing users to freely arrange them according to their preferences.

Prototyping Questions

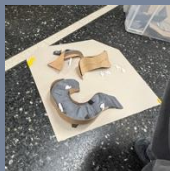
1. Do users prefer the **loop structure**, the **path structure**, or a **combination of both**?

2. What should the design improve based on **hygiene issues** of different **weather conditions**?

3. Where should **seating and table facilities** be positioned to best accommodate both individual and group use while ensuring accessible circulation?

4. Anything you want to add to make it more **playful**?

Users’ Comments



8 out of 12 users indicated a preference for the path structure itself, noting that its form offers a balance of **privacy and social connectivity**. In contrast, the loop structure was perceived as more **isolating**, with users on the upper level disconnected from those below.

“ You should add a **drainage system**, otherwise, rainwater will just run down the slope and form a little puddle.” – Curtis, DE1

“ You should add a **floor** to the path, so we’re not sitting directly on the grass when it’s dirty or wet.” – Tony, ChemEng1



“ I ’d like seats on top of the loop for **better views**.” – Daisy, MBE1



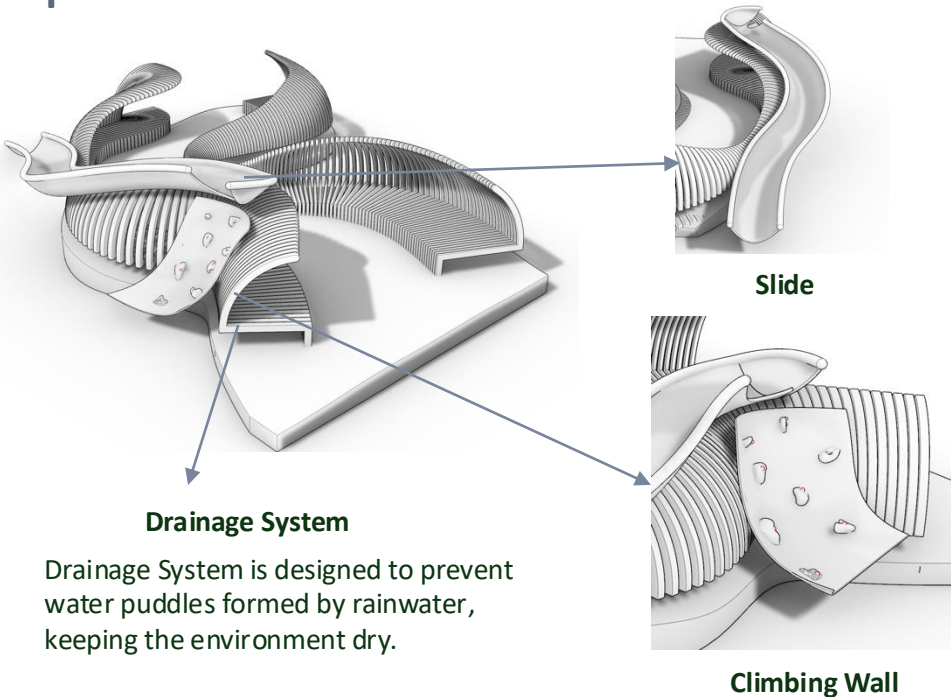
“ I like having the tables and seats inside the round structure—it feels more **private**.” – Joe, DE1

“ A **slide** would be **interesting** and can be built along the shape of the structure.” – Josh
“ It might be fun to add a **climbing wall**—that would be pretty cool!”– Allen, ME

Things to improve

- 1. Drainage System:** A drainage system should be incorporated to prevent the accumulation of rainwater and maintain a clean, organized environment, especially during rainy seasons.
- 2. Flooring Installation:** Adding flooring can prevent users from sitting directly on the grass, enhancing both cleanliness and user comfort while improving accessibility in muddy or uneven areas.
- 3. Playful Elements:** Integrating playful features such as a slide or climbing wall can make the space more engaging and enjoyable, encouraging spontaneous interactions and informal physical activity.
- 4. Structure Preference:** The path design is preferred over a loop configuration, as it strikes a balance between privacy and social interaction while offering more intuitive movement across the space.

Prototype Iteration – Arrangement of Components



7. Specifications & Benchmarking

Specification 1: Materials

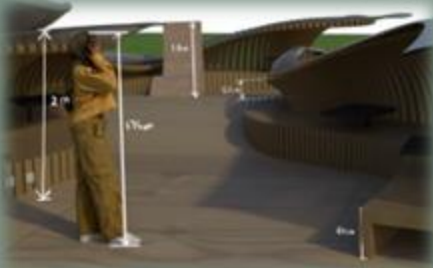
Recycled Plastic Composite is chosen for the Modular seats due to its **waterproof**, **low-maintenance**, and **eco-friendly** properties, offering strong **durability** and **weather resistance** for long-term **outdoor use**.



Polycarbonate is chosen for the transparent part of the chair to allow **light transmission** while ensuring **durability**. Its **high impact resistance** and **weather performance** make it ideal for **outdoor transparent covers**.



Specification 2: Dimensions



- The curved side wall of the structure reaches a maximum height of **2.0 meters**, with the lowest point measuring **0.5 meters**.
- The **slide with the integrated climbing wall** has a total height of **2.2 meters**.
- The **modular chair**, when fully extended, has a height of **1.3 meters**.
- The **seat height** of the chair is designed to be **40 centimeters**, providing ergonomic comfort.

Benchmarking: Compare to other playgrounds & public areas

Public Area



Dalby Court (Imperial College London)



Ziqing Pavilion (Tsinghua University)



Our design

Adaptability to different weather conditions

It is **not** weather-resistant at all. There is **no waterproof protection** or shelter. The functionality is strongly **restricted**.

It can handle **most weather** as it has a rooftop, enabling students to rest on **rainy days**, but it may be a little **moist** during rainy seasons.

The design demonstrates **good adaptability** to varying weather conditions. The **sheltered structure** ensures that activities can take place comfortably during both **rainy** and **sunny** days.

Capacity & Comfort

The capacity is restricted to a maximum **2-3 users simultaneously** for a single unit.

It is **not ergonomically** designed, hence users will **not feel comfortable** when sitting for a long time on the bench.

The pavilion is able to hold **multiple users simultaneously**.

The seatings are made out of stones, hence the seating experience is **not that comfortable**.

It **can hold multiple users** due to the multiple seating and sufficient spacing of the design.

It is **perfectly ergonomically** designed to satisfy the users' needs. Due to user feedback, the seating experience is very **comfortable**.

Culture & Aesthetic

It is **poorly** designed for aesthetics, and it is **not linked** to school culture at all.

It has a **perfect appearance** and it is strongly associated with the school culture, doing a great promotion for the school.

The aesthetic outlook is good, and the **school culture is linked** by setting up **screens** in the design to display school cultural contents.

Multifunctionality

It provides space for students to **study, chat, and have informal meetings**.

The lack of tables limits the space for **seating and chatting**, reducing its **usability for study**.

The design includes **tables and chairs** to support both **relaxation and study**, while the **modular seating** adds a playful element. **Slides** are also integrated to enhance **entertainment** and engagement.

8. Final Design

Due to the limitations of making a large structure, the final design is created and rendered by CAD models.



Screen



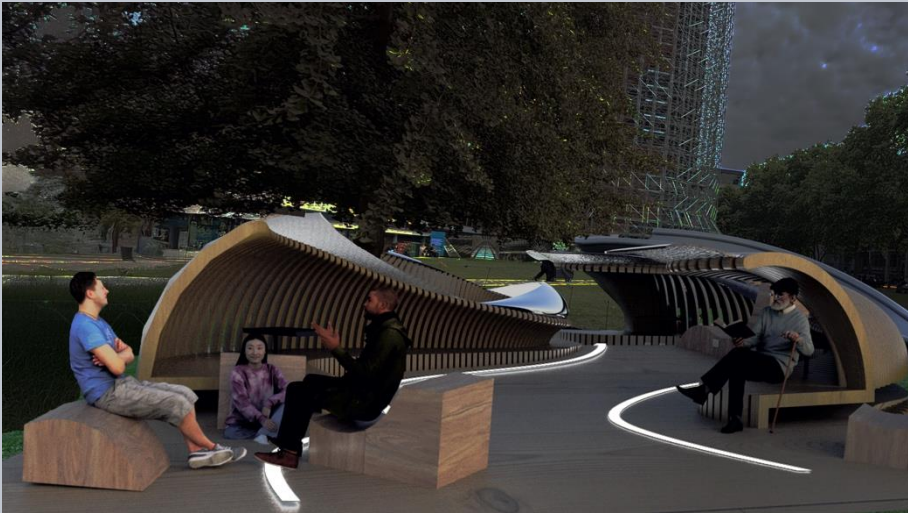
People using the slide



Front View



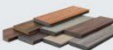
Back View



Night View

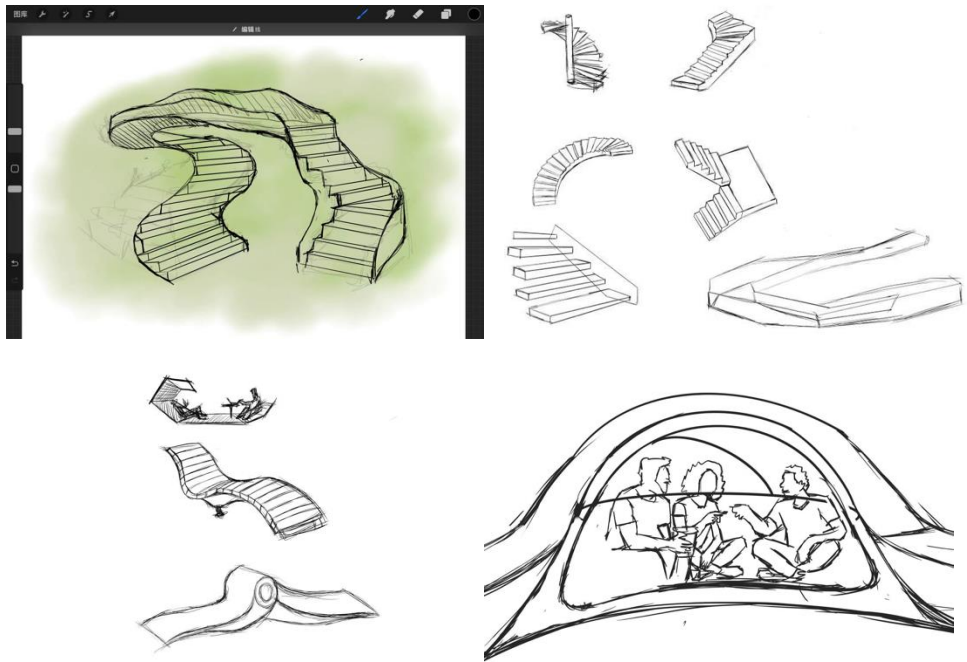
Appendix

Tables of material selection

Material	Appearance	Key Strengths	Key Weaknesses
HDPE (High-Density Polyethylene)		Waterproof, resistant to corrosion, lightweight	Can warp or expand slightly under extreme heat, less stable
Recycled Plastic Composite		Waterproof, low maintenance, eco-friendly	Low stiffness, low thermal resistance.
GFRC (Glass Fiber Reinforced Concrete)		Very Strong, Durable, weatherproof	Heavy, Brittle, Expensive
Teak Wood		Naturally weather-resistant, resists rot and insects, ages beautifully	Expensive, Heavy, Require maintenance.
WPC – Wood Plastic Composite		Looks like wood, performs like plastic; waterproof and insect-proof	Heavy, Low tensile strength

Material	Transparency	Durability	Weight	UV Resistance	Cost	Best Use
Tempered Glass	High	Good	Heavy	✓✓	High	Premium canopy, flat or slightly curved
Laminated Glass	High	Good	Heavy	✓✓✓	Very High	Safety-critical, public roofing
Polycarbonate	High	Good	Low	✓✓✓	Medium	Transparent, durable cover

Previous sketches

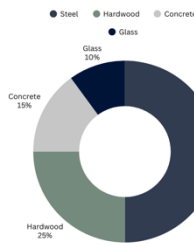


Software used:

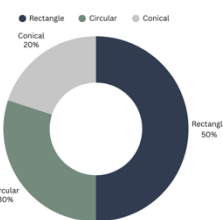
- Fusion 360 for CAD model
- Rhino for CAD model
- Procreate for sketch drawing
- Adobe Photoshop for photo editing
- Adobe Illustrator
- Adobe Indesign

Design Analysis

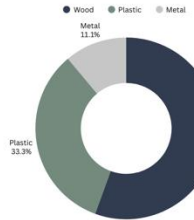
Exterior Frame Materials



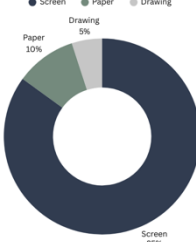
Exterior Frame Shapes



Materials for chairs



Information Display



Unused CAD Models made during Prototype

